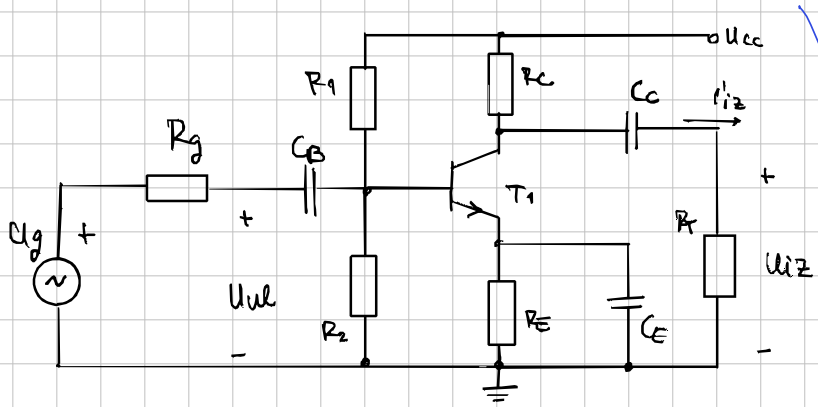
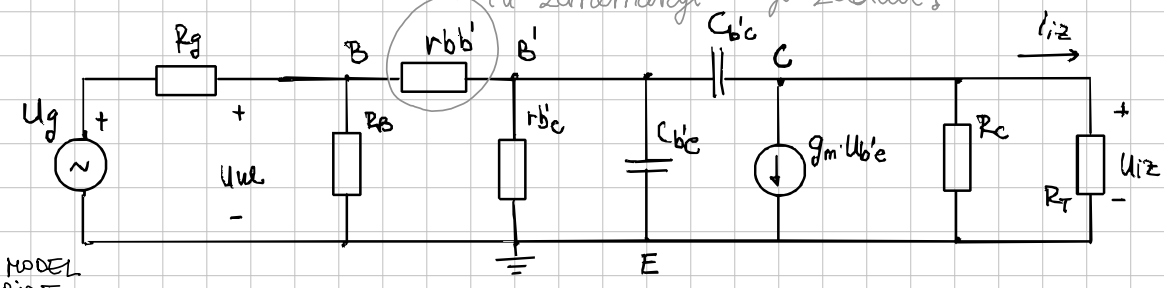
SZE - visoke frekvencije

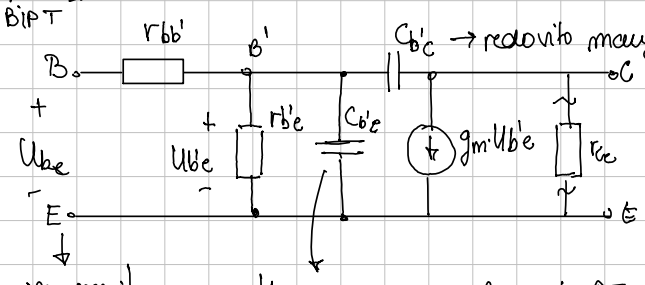


kreećemo od niskih i idemo na više frekv.

ili zanemarimo ili je zadan!



MODEL BIPT



$C_{b'c} \rightarrow$ redovito manji od $C_{b'c}$ $\Rightarrow$ jer je kapacitet nepropusno polariziranog spoja BC

na masi!

Kap. propusno pol. spoja BE
$\Rightarrow$ dosta velik kapacitet
$\Rightarrow$ difuzijski kapacitet

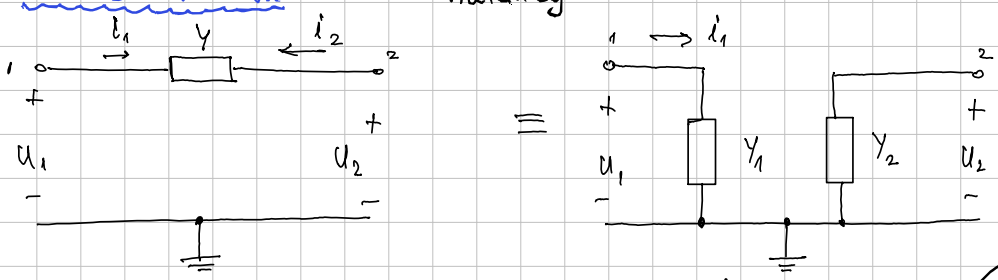
$\rightarrow$ kapacitet $C_{b'c}$ povezuje izlaz i ulaz
? postoji li način pojednostavljivanja

$\Downarrow$

MILLEROV TM

Millerov teorem

admitancije



originalna mreža — transformirana mreža

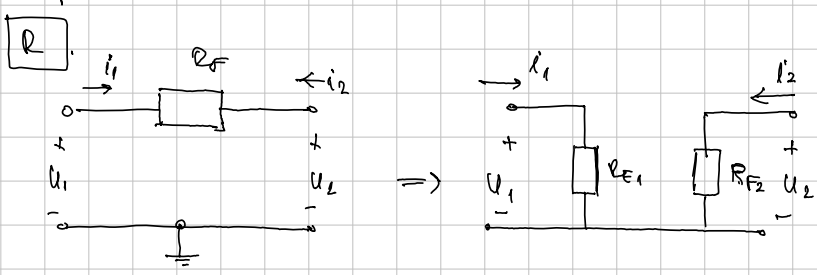
$$K = \frac{U_2}{U_1}$$

$$I_1 = Y(U_1 - U_2) = YU_1(1-K) = Y_1 U_1 \longrightarrow Y_1 = Y(1-K)$$

$$I_2 = Y(U_2 - U_1) = YU_2\left(1-\frac{1}{K}\right) = Y_2 U_2 \longrightarrow Y_2 = Y\left(1-\frac{1}{K}\right) = Y\frac{K-1}{K}$$

impedancije: $Z_1 = \dfrac{Z}{1-K}$ $\quad Z_2 = Z\dfrac{K}{K-1}$

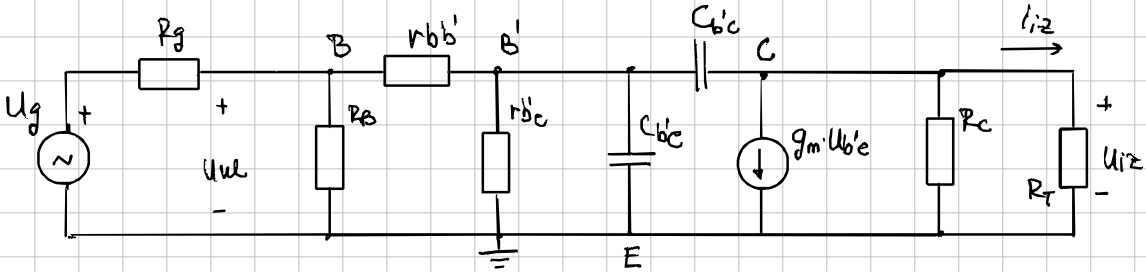
Npr:



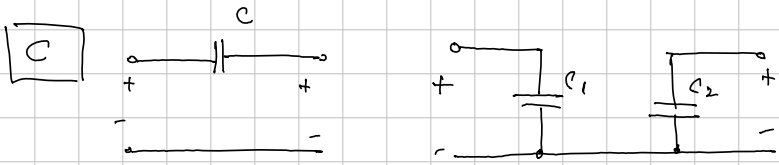
$$Z_1 = R_{F1} = \frac{R_F}{1-K}$$

$$Z_2 = R_{F2} = R_F \cdot \frac{K}{K-1}$$

Primjena Millerovog teorema:



Kako kapacitet:



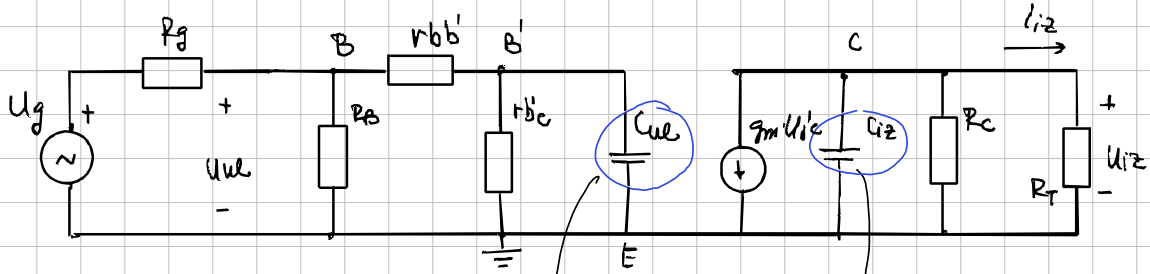
$$C_1: Z_1 = \frac{Z}{1-K} = \left(\frac{1}{j\omega C}\right)\cdot\frac{1}{1-K} = \frac{1}{j\omega \underbrace{C(1-K)}_{C_1}}$$

$$C_2: Z_2 = \frac{1}{j\omega C}\cdot\frac{K}{K-1} = \frac{1}{j\omega \underbrace{C\frac{K-1}{K}}_{C_2}}$$

→ znači dio će se preslikati na ulaz, a dio na izlaz

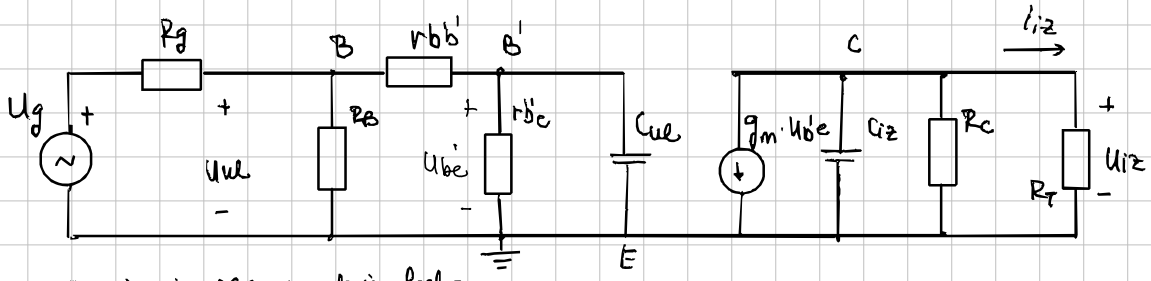
↓

preslikani: $C_{b'c} \rightarrow C_{b'c,ul} = C_{b'c}(1-K)$ $\qquad C_{b'c,iz} = C_{b'c}\cdot\frac{K-1}{K}$



$$C_{ul} = C_{b'c} + C_{b'c}(1-K) \qquad C_{iz} = C_{b'c}\frac{K-1}{K}$$

→ radimo analizu sklopa na srednjim i na visokim frekv.



→ zanimaju nas srednje frekv i visoke frekv

Srednje frekvencije → sri kond od linije signala prema masi → polovi
→ raste pojačanje na visokim f. (Cul i Ciz) (postaju K.S.)

$A_{vo} = ?$

$\boxed{\omega \to \infty}$, $Z_C = \frac{1}{j\omega C}$; $Z_C \to 0$ $(u_{uc} \to 0)$ $(g_m \cdot u_{b'e} \to 0)$ => $u_{iz} \to 0$

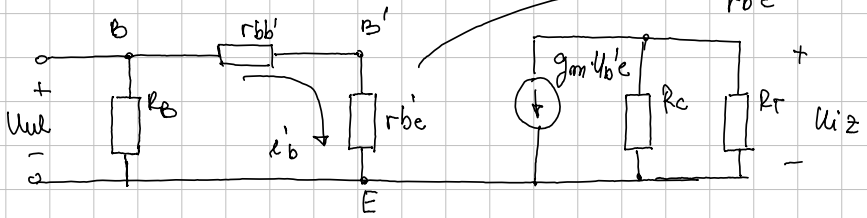
$A_{vo} = 0$

$\boxed{\omega \to 0}$: VISOKE FREKVENCIJE

→ tj gledamo da frekv teže prema srednjim frekv

→ C_E, C_C, C_B → postali K.S.

→ Srednje frekvencije



$i_b = \frac{u_{b'e}}{r_{b'e}}$

• na srednjim frekv. Cul i Ciz se odspajaju (jer još nisu počeli djelovati)

$$A_{vo} = \frac{u_{iz}}{u_{ul}} = \frac{-g_m \cdot \cancel{u_{b'e}} (R_C \parallel R_T)}{\frac{\cancel{u_{b'e}}}{r_{b'e}} \cdot (r_{bb'} + r_{b'e})} = -g_m \cdot r_{b'e} \frac{(R_C \parallel R_T)}{r_{bb'} + r_{b'e}} \approx -g_m \cdot (R_C \parallel R_T)$$

$r_{b'e} \approx 1k\Omega$, $r_{bb'} \approx 10\Omega$

↳ POJAČANJE NA SREDNJIM FREKV.

$\boxed{C_{ul}}$ → Ciz = odspojiti $A_v' = \frac{u_{iz}}{u_{b'e}} \Rightarrow \frac{u_{iz}}{u_{b'e}} \cdot \frac{u_{b'e}}{u_{ul}} = A_v(j\omega)$

$$\boxed{A_v' = \frac{-g_m \cdot \cancel{u_{b'e}} (R_C \parallel R_T)}{\cancel{u_{b'e}}} = -g_m (R_C \parallel R_T)}$$

→ nema frekv. ovisnosti

→ vidimo je i K iz Millerovog TM

K = omjer napona s jedne i druge str kondenzatora koji smo preslikali

$A_v' = K = \frac{u_{iz}}{u_{b'e}} < 0$

$C_{b'c,ul} = C_{b'e}(1-K) = C_{b'e}(1 + g_m (R_C \parallel R_T))$

↳ pojačanje na srednjim frekvencijama

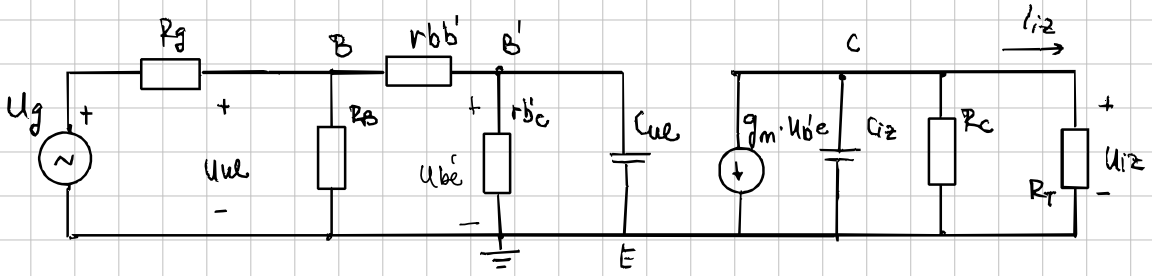
↳ kapacitet preslikana u ulazni krug

ondiko puta veći koliko je pojačanje ovog pojačala na srednjim frekv

→ originalni je ≈ 10pF i odjednom se preslikava 100 puta veći => LOŠE

$$C_{ul} = C_{b'e} + C_{b'c}(1 + g_m (R_C \parallel R_T))$$

postaju usporedivi



Znači $A_v(j\omega) = \frac{u_{iz}}{u_{b'e}} \cdot \frac{u_{b'e}}{u_{ul}} = \underbrace{A_v'}_{\text{ovo smo prije izrazili}} \cdot \frac{u_{b'e}}{u_{ul}}$

$\frac{u_{b'e}}{u_{ul}} = ?$

$$i_b = \frac{u_{ul}}{r_{bb'} + r_{b'e} \parallel \frac{1}{j\omega C_{ul}}}$$

$$u_{b'e} = i_b \cdot r_{b'e} \parallel \frac{1}{j\omega C_{ul}} = \frac{u_{ul}\left(r_{b'e} \parallel \frac{1}{j\omega C_{ul}}\right)}{r_{bb'} + r_{b'e} \parallel \frac{1}{j\omega C_{ul}}}$$

$$= u_{ul} \frac{r_{b'e} \cdot \frac{1}{j\omega C_{ul}}}{r_{b'e} + \frac{1}{j\omega C_{ul}}} \cdot \frac{1}{r_{bb'} + \frac{r_{b'e} \cdot \frac{1}{j\omega C_{ul}}}{r_{b'e} + \frac{1}{j\omega C_{ul}}}}$$

$$\frac{u_{b'e}}{u_{ul}} = \frac{r_{b'e} \cdot \frac{1}{j\omega C_{ul}}}{r_{bb'}\left(r_{b'e} + \frac{1}{j\omega C_{ul}}\right) + r_{b'e} \cdot \frac{1}{j\omega C_{ul}}} = \frac{r_{b'e}}{r_{bb'}(r_{b'e} \cdot j\omega C_{ul} + 1) + r_{b'e}}$$

$$= \frac{r_{b'e}}{r_{bb'} \cdot r_{b'e} \cdot j\omega C_{ul} + r_{bb'} + r_{b'e}} = \frac{r_{b'e}}{r_{bb'} + r_{b'e}} \cdot \frac{1}{1 + j\omega C_{ul} \cdot \frac{r_{bb'} \cdot r_{b'e}}{r_{bb'} + r_{b'e}}}$$

$$\frac{u_{b'e}}{u_{ul}} = \frac{r_{b'e}}{r_{bb'} + r_{b'e}} \cdot \frac{1}{1 + j\omega C_{ul} \cdot (r_{bb'} \parallel r_{b'e})}$$

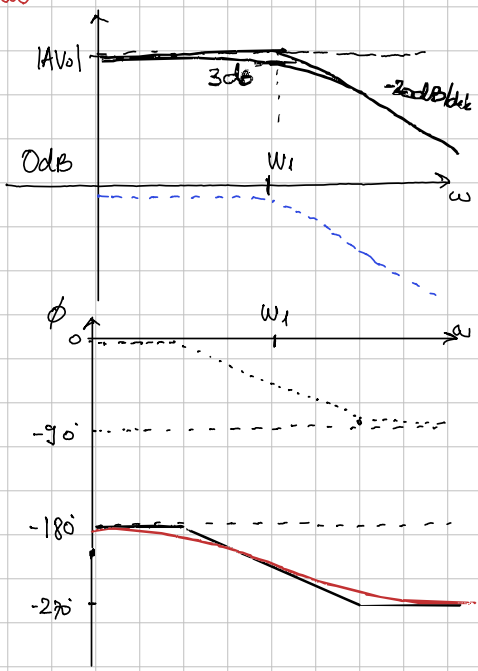
$$\Rightarrow A_v(j\omega) = \underbrace{\underbrace{-g_m \cdot (R_C \parallel R_T)}_{A_v'} \cdot \frac{r_{b'e}}{r_{bb'} + r_{b'e}}}_{A_{v0}\ \text{realan član}} \cdot \underbrace{\frac{1}{1 + j\omega C_{ul} \cdot (r_{bb'} \parallel r_{b'e})}}_{\text{frekvencijski ovisan član}}$$

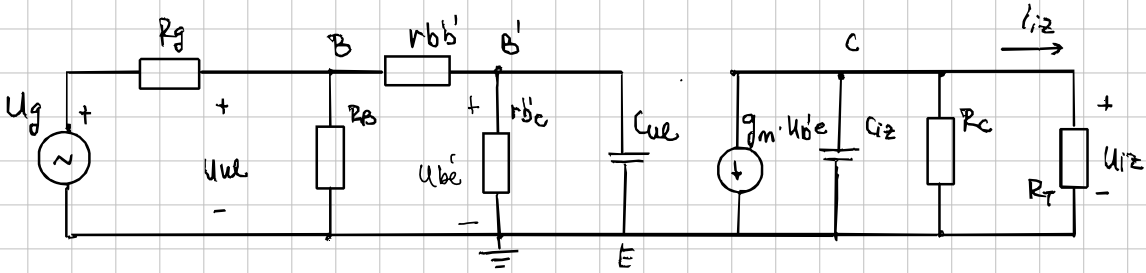
1. na vrlo niskim frekv (frekv. dio ne djeluje) → A_{v0} je realan dio ⇒ A_{v0}
odnosno frekv na kojima ne djeluje taj dio

$$A_v(j\omega) = A_{v0} \cdot \frac{1}{1 + j\frac{\omega}{\omega_1}}$$

$$\omega_1 = \frac{1}{T_1} = \frac{1}{C_{ul}(r_{bb'} \parallel r_{b'e})}$$

karakt visoko prop filtra
- niske frekv propušta
- kako frekv raste, karakt pada





Možemo li samo gledajući shemu procijeniti / dobiti ω_1 ?

PROCJENA: Što vidi Cul

$$\omega_1 = \frac{1}{C_{ul} \cdot r_{b'e} \parallel (r_{bb'})}$$

✓ jer Uul K.S. pa ne vidi RB

→ ne treba nam ovako dug izvod

$\boxed{C_{iz}}$ → „taktika gledanja" pretpostavljamo da ga gledamo kao pol

$$\cdots \frac{1}{1 + j\frac{\omega}{\omega_2}}$$

$$\omega_2 = \frac{1}{\tau_2} = \frac{1}{C_{iz} \cdot (R_C \parallel R_T)}$$

✓ imamo li otpor preko strujnog izvora ?

→ Uul = K.S. ⟹ svi naponi i sve struje su Ø

⟹ $U_{b'e} = 0$

⟹ $g_m \cdot U_{b'e} = Ø$

NEMA GA

$$\boxed{A_v(j\omega) = A_{v0} \cdot \frac{1}{1 + j\frac{\omega}{\omega_1}} \cdot \frac{1}{1 + j\frac{\omega}{\omega_2}}}$$

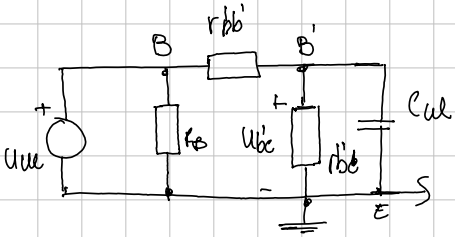
$$C_{iz} = C_{b'c} \frac{K-1}{K}$$

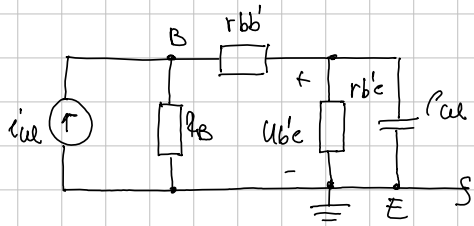
$$K = -g_m (R_C \parallel R_T)$$

$$C_{iz} = C_{b'c} \frac{1/g_m (R_C \parallel R_T) + 1}{1/g_m (R_C \parallel R_T)} \approx \frac{101}{100} \Rightarrow \boxed{C_{iz} \approx C_{b'c}}$$

nemamo problema o preslikavanju

Vremenska konstanta τ_{ul} (skraćeno)





Za naponsko pojačanje

$$\tau_{ul} = (r_{bb'} \| r_{b'e})\, C_{ul}$$

Za strujno pojačanje

$$\tau_{ul} = \left[(R_B + r_{bb'}) \| r_{b'e} \right] C_{ul}$$

uz konačni R_g $\quad \tau_{ul} = \left[(R_g \| R_B + r_{bb'}) \| r_{b'e} \right] C_{ul}$

Millerov efekt

$$C_{ul} = C_{b'e} + C_{b'c}\left[1 + g_m (R_C \| R_T)\right]$$

Kapacitet $C_{b'c}$ spojen između izlaza i ulaza preslikava se u ulazni krug pomnožen s iznosom naponskog pojačanja → smanjenje gornje granične f.